Lesson 1: Squares, Square Roots and Triangle Basics

Squaring a number, finding and estimating square roots, the three angles of a triangle, naming triangles by their sides and angles, and the parts of a right triangle.

1. Squares and Perfect Squares

To **square** a number, multiply it by itself. We write a small 2 up high: 6^2 is read "6 squared" and means $6 \times 6 = 36$. The small 2 is called an **exponent**.

Why "squared"? A square that is 6 units long on each side is covered by $6 \times 6 = 36$ small squares. So the **area of a square** is its side length squared.

A **perfect square** is what you get when you square a whole number: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, ... So 1, 4, 9 and 16 are perfect squares.

Squaring: $n^2 = n \times n$	$6^2 = 6 \times 6 = 36$	**Area of a square** $= (\text{side})^2$

Watch out. 6^2 means $6 \times 6 = 36$. It does **not** mean $6 \times 2 = 12$. The exponent tells you how many 6s to multiply.

Example 1

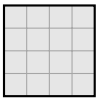
(a) Each small square is 1 square unit. Write the area of each big square as a square number and as a number.



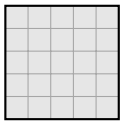
Side 2: $2^2 =$ _____



Side 3: $3^2 =$ _____



Side 4: $4^2 =$ _____



Side 5: $5^2 =$ _____

(b) Complete the table of perfect squares. Keep it: you will use it all unit.

Number n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Square n^2	1	4													

Calculator. To square a number, type the number, press x^2, then ENTER. For 2.4^2: 2.4 x^2 ENTER shows `5.76`.

Example 2

Find each square. Use your calculator for (d) to (f).

(a) 9^2

(b) 12^2

(c) 20^2

(d) 1.5^2

(e) 0.3^2

(f) 2.4^2

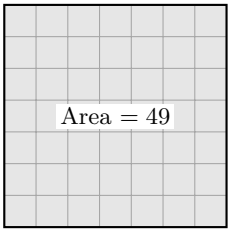
(g) Jaspreet is laying a square patio that is 3.5 m long on each side. What is the area of the patio?

Did you know? When you square a number less than 1, the answer gets *smaller*: $0.3^2 = 0.09$. You are taking 0.3 of something that is already small.

2. Square Roots

Finding a **square root** is the opposite of squaring. It "undoes" the square. The square root of 49 is 7, because $7^2 = 49$. We write $\sqrt{49} = 7$. The sign $\sqrt{\ }$ is the **square root sign**.

If you know the **area** of a square, its **side length** is the square root of the area.



side $= \sqrt{49} = 7$

square: side 7 $\longrightarrow$ area $7^2 = 49$

square root: area 49 $\longrightarrow$ side $\sqrt{49} = 7$

$\sqrt{49} = 7$ because $7^2 = 49$. **Side of a square** $= \sqrt{\text{area}}$

A **perfect square** has a whole-number square root ($\sqrt{81} = 9$). Any other whole number has a square root that is a decimal that never ends ($\sqrt{40} = 6.324\,555\ldots$). Round it as the question asks, often to the nearest tenth.

Calculator. The $\sqrt{\ }$ key is above x^2. Press 2nd x^2. The screen shows $\sqrt{(}$. Type the number, then) and ENTER.
For $\sqrt{40}$: 2nd x^2 40) ENTER shows `6.32455532`.

Watch out. A square root is **not** half the number: $\sqrt{64}$ is 8, not 32. Check any square root by squaring it: $8^2 = 64$ ✓, but $32^2 = 1024$.

Example 3

(a) Find each square root without a calculator. Use the table of perfect squares.

$\sqrt{36} =$ _____ $\sqrt{100} =$ _____ $\sqrt{169} =$ _____ $\sqrt{1} =$ _____

(b) Use your calculator. Round to the nearest tenth. Then check by squaring your answer.

$\sqrt{40}$ $\sqrt{12}$ $\sqrt{150}$ $\sqrt{6.5}$

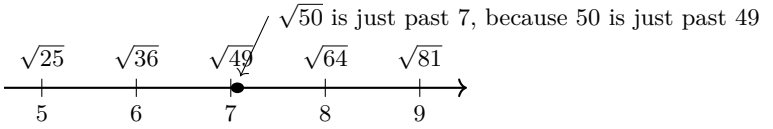
Example 4

(a) A square vegetable garden has an area of 50 m^2. How long is each side, to the nearest tenth of a metre? How much fencing goes all the way around it?

(b) A square rug covers an area of 5.76 m^2. How long is each side of the rug?

3. Estimating a Square Root

Before you use a calculator, **estimate**. Then you can tell if the calculator answer makes sense. The trick: find the two perfect squares that the number sits between.



To estimate $\sqrt{50}$:
Step 1. Find the perfect squares just below and just above the number: $49 < 50 < 64$.
Step 2. Take their square roots: $\sqrt{49} = 7$ and $\sqrt{64} = 8$. So $\sqrt{50}$ is **between 7 and 8**.
Step 3. Which perfect square is the number closer to? 50 is much closer to 49, so $\sqrt{50}$ is **a little more than 7**.

Example 5

Estimate each square root. Then use your calculator to check (round to the nearest tenth).

Number	Between which perfect squares?	So the root is between	Closer to	Calculator
$\sqrt{27}$				
$\sqrt{62}$				
$\sqrt{105}$				
$\sqrt{8}$				

Example 6

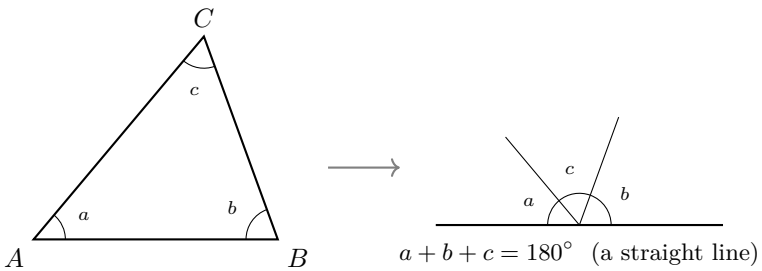
(a) Ty pressed some keys and wrote $\sqrt{90} = 45$. Is his answer reasonable? Explain, and find the right answer.

(b) A square window has an area of 1.2 m^2. Kara says each side is about 1.1 m. Is that reasonable?

4. Triangles: The Angle Sum, and Naming Triangles

A **triangle** has 3 sides, 3 corners and 3 angles. Each corner is a **vertex** (plural: vertices) and is named with a capital letter. The triangle with vertices A, B and C is written $\triangle ABC$.

If you tear the three corners off any paper triangle and put them side by side, they always make a straight line. A straight line is 180° (Unit 5).



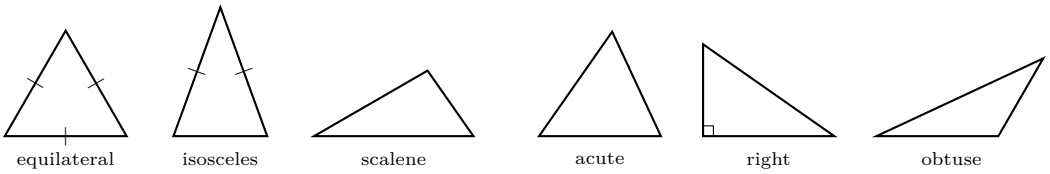
The angles in a triangle add up to $180°$**:** $\angle A + \angle B + \angle C = 180°$

Missing angle $= 180° -$ (the two angles you know).

In a **right triangle**, the right angle uses $90°$, so the other two angles add up to $90°$ (they are complementary).

Triangles get names from their **sides** and from their **angles**. Small tick marks show sides that are the same length.

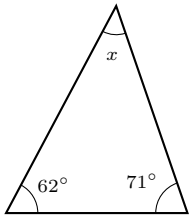
By sides	**By angles**
Equilateral: all 3 sides equal. All 3 angles are $60°$.	**Acute:** all 3 angles are less than $90°$.
Isosceles: 2 sides equal. The 2 angles across from the equal sides are also equal.	**Right:** one angle is exactly $90°$ (marked with a small square).
Scalene: no sides equal, and no angles equal.	**Obtuse:** one angle is more than $90°$.

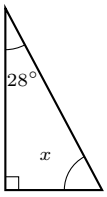


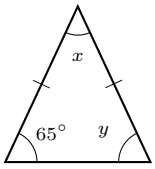
Watch out. Every triangle has at least two acute angles. The **biggest** angle decides the name: if it is $90°$ the triangle is right, if it is more than $90°$ it is obtuse. A triangle can never have two right angles or two obtuse angles (they would already use up $180°$ or more). A triangle gets one name of each kind, like a **right isosceles** triangle.

Example 7

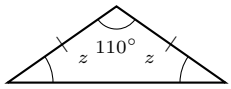
Find each missing angle. Write the equation first.

(a)


(b)


(c)


(d) A roof truss is an isosceles triangle. The angle at the peak is $110°$. What are the two equal angles at the bottom?



Example 8

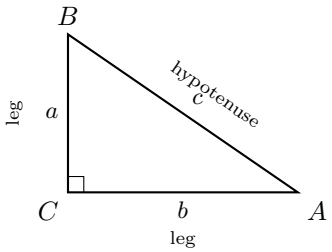
Name each triangle by its sides and by its angles.

	Side lengths	Angles	By sides	By angles
(a)	5 cm, 5 cm, 5 cm	60°, 60°, 60°		
(b)	6 cm, 6 cm, 8.5 cm	45°, 45°, 90°		
(c)	4 cm, 7 cm, 9 cm	25°, 48°, 107°		
(d)	6 cm, 8 cm, 10 cm	37°, 53°, 90°		
(e)	7 cm, 7 cm, 4.8 cm	70°, 70°, 40°		

5. The Parts of a Right Triangle

A **right triangle** has one right angle (90°), shown by a small square in the corner. Its sides have special names:

- The two sides that make the right angle are the **legs**.
- The side across from (**opposite**) the right angle is the **hypotenuse** (say "hy-POT-en-use"). It is always the **longest** side.



Naming the sides of $\triangle ABC$

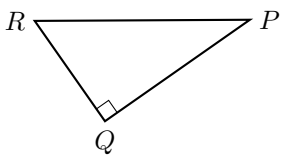
1. With the **small letter** of the vertex across from it: side a is across from $\angle A$, side b across from $\angle B$, side c across from $\angle C$.

2. With its **two end points**: side a is also side BC.

In this unit the right angle is usually at C. Then the **hypotenuse is** c (side AB), and the **legs are** a **and** b.

Tip. To find the hypotenuse, put your finger on the little square and slide straight across the triangle. The side you land on is the hypotenuse. A triangle can be turned any way, so the hypotenuse is not always the slanted side or the bottom side.

Example 9



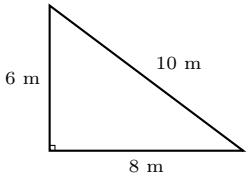
$\triangle PQR$ has its right angle at Q.

(a) Label each side with a small letter.

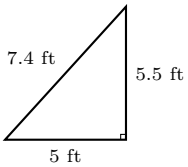
(b) The hypotenuse is side _____, also called side _____.

(c) The legs are sides _____ and _____, also called _____ and _____.

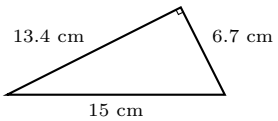
(d) Write the length of the hypotenuse of each right triangle. Check that it is the longest side.



Hypotenuse: _____



Hypotenuse: _____



Hypotenuse: _____

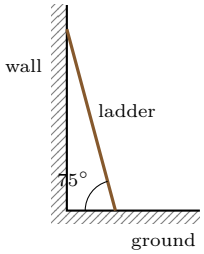
Example 10

A ladder leans against the wall of a house. The ground is level and the wall goes straight up.

(a) Where is the right angle? Mark it on the picture.

(b) Which part is the hypotenuse? Which parts are the legs?

(c) The ladder makes a 75° angle with the ground. What angle does it make with the wall?



Coming next. In Lesson 2 you will discover how the three sides of a right triangle are connected. You will need squares and square roots from this lesson, so keep your table of perfect squares handy.

Lesson 2: The Pythagorean Theorem: Finding the Hypotenuse

Discovering how the three sides of a right triangle are connected, using $a^2 + b^2 = c^2$ to find the longest side, seeing why it only works for right triangles, and solving real problems with it.

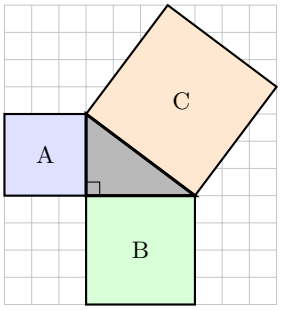
1. Squares on the Sides of a Right Triangle

A **right triangle** has one right angle (90°), shown with a small square in the corner. The two sides that make the right angle are the **legs**. The side across from the right angle is the **hypotenuse**. The hypotenuse is always the **longest** side.

Draw a square on each side of a right triangle. The areas of the squares are connected in a surprising way. Let's find out how.

Example 1

Each small square on the grid is 1 square unit. The legs of the right triangle are 3 units and 4 units long.



(a) Count the squares. Area of square A (on the 3-unit leg): _____

Area of square B (on the 4-unit leg): _____

(b) Square C sits on the hypotenuse. It is tilted, so it is hard to count. Draw a 7×7 box around it (corner at the right angle). The box has area _____. Inside the box but outside square C are 4 triangles (one is the grey one). Each has legs 3 and 4, so each has area $3 \times 4 \div 2 =$ _____.

Area of square C = _____ − _____ = _____

(c) Compare: A + B = _____. The side of square C is _____ units, so the hypotenuse is 5 units long.

Example 2

A class drew three more right triangles on grid paper and measured the sides. Complete the table.

Leg a	**Leg** b	**Hypotenuse** c	a^2	b^2	$a^2 + b^2$	c^2
6 cm	8 cm	10 cm				
5 cm	12 cm	13 cm				
8 cm	15 cm	17 cm				

What do you notice? $a^2 + b^2$ is always _____.

2. The Pythagorean Theorem

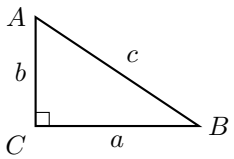
The pattern you found is called the **Pythagorean theorem**. A **theorem** is a math rule that has been proved to be true every time.

The Pythagorean theorem. In a **right** triangle with legs a and b and hypotenuse c:

$$a^2 + b^2 = c^2$$

In words: $(\text{leg})^2 + (\text{leg})^2 = (\text{hypotenuse})^2$.

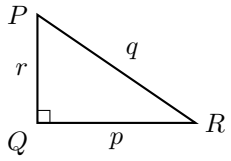
The square on the hypotenuse has the same area as the two squares on the legs put together.



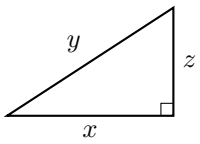
Naming the sides. Each corner (vertex) gets a capital letter. Each side gets the small letter of the corner *across from it*: side a is across from $\angle A$, and so on. We usually put the right angle at C, so the hypotenuse is c. If a triangle uses other letters, find the right angle first: the side across from it is the hypotenuse, and its square goes alone on one side of the equals sign.

Example 3

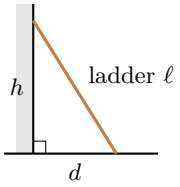
For each right triangle, name the hypotenuse. Then write the Pythagorean theorem using the triangle's own letters.



(a) Hypotenuse: _____



(b) Hypotenuse: _____



(c) A ladder leans on a wall. Hypotenuse: _____

Did you know? The theorem is named after **Pythagoras**, a Greek thinker who lived about 2500 years ago. But people knew the pattern long before him. A clay tablet from ancient **Babylon** (in today's Iraq), about 3800 years old, lists sets of numbers that fit $a^2 + b^2 = c^2$. Ancient writings from India and China also describe the rule. **Today** the theorem is used to find the size of TV and phone screens, the length of ramps, stairs and roof rafters, the straight-line distance on a map, and in video games to work out how far apart two players are. Carpenters use it to check that corners are square (next lesson).

3. Finding the Hypotenuse

If you know both legs, you can find the hypotenuse. Square each leg, add, then take the **square root** to undo the squaring.

To find the hypotenuse c:
Step 1. Find the right angle. Label the legs a and b and the hypotenuse c.
Step 2. Write $a^2 + b^2 = c^2$ and substitute the two legs.
Step 3. Square each leg, then add.
Step 4. Take the square root: $c = \sqrt{a^2 + b^2}$.
Step 5. Round, and check: c must be **longer** than each leg but **shorter** than the two legs added together.

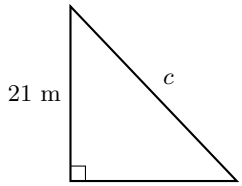
Calculator. To square a number, press the number then x^2. The square root $\sqrt{\ }$ is 2nd x^2. You can do it all at once: 2nd x^2 7.5 x^2 + 4 x^2) ENTER gives 8.5. The $\sqrt{\ }$ opens a bracket, so close it with) before ENTER.

Watch out. Do **not** just add the legs: $7.5 + 4 = 11.5$ is too long. And don't forget the last step: $c^2 = 72.25$ is not the side length. You still need $\sqrt{72.25} = 8.5$.

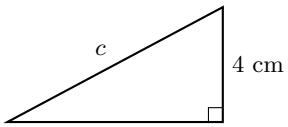
Example 4

Find the length of the hypotenuse. Round to the nearest tenth if needed. Check that each answer is reasonable.

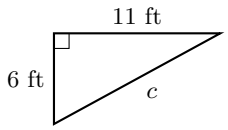
(a)



(b)



(c)

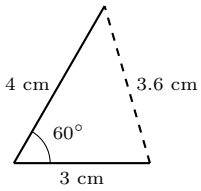


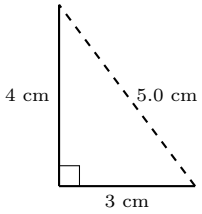
4. Why It Only Works for Right Triangles

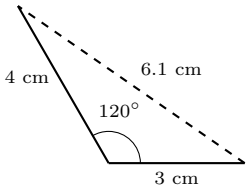
The Pythagorean theorem is **only** true for right triangles. One example where a rule fails is called a **counterexample**. One counterexample is enough to show a rule does not work for every triangle.

Example 5

Picture two sticks, 3 cm and 4 cm long, joined at one end by a hinge. Open the hinge to three different angles and measure the distance c between the free ends.







Angle at hinge	**Third side** c	c^2	$3^2 + 4^2$	**Is** $3^2 + 4^2 = c^2$**?**
60°	3.6 cm			
90°	5.0 cm			
120°	6.1 cm			

Complete: When the angle is smaller than 90°, c^2 is _____ $a^2 + b^2$. When it is bigger than 90°, c^2 is _____ $a^2 + b^2$. They are equal only when the angle is _____.

Try it. Cut two paper strips, or use a free geometry app on a computer. Make a triangle, measure the three sides, and work out $a^2 + b^2$ and c^2. Drag one corner and watch: the two numbers match only when the angle is exactly 90°.

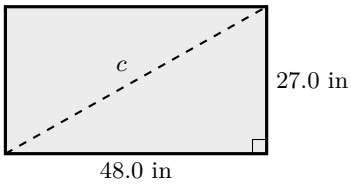
5. Solving Problems with Right Triangles

Right triangles are everywhere: a wall meets flat ground at 90°, the corner of a rectangle (a screen, a field, a door) is 90°, and north meets east at 90°. When a problem asks for a slanted or diagonal distance, look for the right triangle.

Solving a problem: (1) Sketch it and mark the right angle. (2) Label the two legs and the hypotenuse; make sure all lengths are in the **same unit**. (3) Use $a^2 + b^2 = c^2$. (4) Check that the answer is reasonable. (5) Answer with a sentence.

Example 6

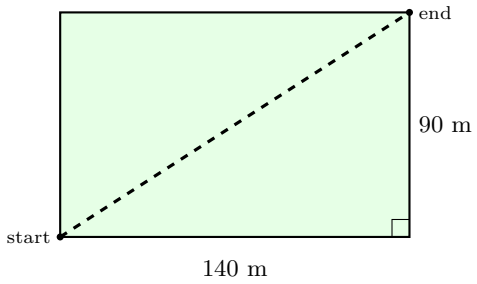
A TV's size is the length of the **diagonal** of its screen, in inches. Amira's new TV screen is 48.0 in wide and 27.0 in high.
(a) Find the length of the diagonal, to the nearest inch.



(b) The box says "55-inch TV". Is that right?

Example 7

A park in Airdrie is a rectangle 140 m long and 90 m wide. Noah usually walks along two edges to get to the opposite corner. Today he cuts straight across the grass.
(a) How far is the walk along the two edges?



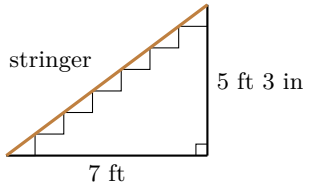
(b) How far is the shortcut, to the nearest tenth of a metre?

(c) How much shorter is the shortcut?

Feet and inches. Before you square, change every length to **one** unit, usually inches (1 ft = 12 in). At the end, change back: divide by 12; the whole number is feet and the remainder is inches.

Example 8

Kenji is building a set of deck stairs. The side board that holds the steps (the **stringer**) is the hypotenuse of a right triangle. The stairs rise 5 ft 3 in and run 7 ft along the ground.
(a) How long must the stringer be? Give your answer in feet and inches.



(b) Boards come in 8 ft, 10 ft and 12 ft lengths. Which should he buy?

Lesson 3: The Pythagorean Theorem: Finding a Leg, and Squaring a Corner

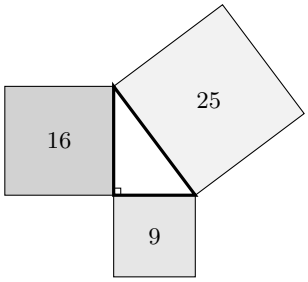
Rearranging the Pythagorean theorem to find a missing leg, testing whether a triangle is a right triangle, and using the 3-4-5 method to make and check square corners.

1. Finding a Missing Leg

In a right triangle, the two sides that make the right angle are the **legs** (a and b). The side across from the right angle is the **hypotenuse** (c). It is always the longest side. The **Pythagorean theorem** says

$$a^2 + b^2 = c^2.$$

In Lesson 2 you knew both legs and found the hypotenuse. Now you know the hypotenuse and *one* leg, and you want the other leg.



The squares on the sides of this right triangle have areas 9, 16 and 25.
The two small squares together make the big one: $9 + 16 = 25$.
So if you take one small square away from the big one, the other small square is left:

$$25 - 16 = 9 \qquad \text{and} \qquad 25 - 9 = 16.$$

To find a leg, **subtract**: (hypotenuse)^2 − (the other leg)^2.

Rearranging the formula. Start with $a^2 + b^2 = c^2$. To get a^2 by itself, subtract b^2 from both sides: $a^2 = c^2 - b^2$.

Missing hypotenuse: $c^2 = a^2 + b^2$ (**add** the squares) **Missing leg:** $a^2 = c^2 - b^2$ or $b^2 = c^2 - a^2$ (**subtract** the squares)

Step 1. Find the hypotenuse: it is across from the right angle. **Step 2.** Write the formula for the side you want. **Step 3.** Substitute and square. **Step 4.** Subtract (or add). **Step 5.** Take the square root. **Step 6.** Check: the hypotenuse is the longest side.

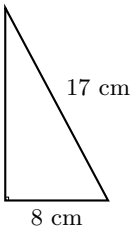
Watch out. When you find a leg, always do **big square minus small square** (hypotenuse^2 first). A leg is always **shorter** than the hypotenuse. If your leg is longer than the hypotenuse, you added when you should have subtracted.

Calculator. On the TI-30XIIS, $\sqrt{\ }$ is the second job of the x^2 key. To find $\sqrt{10.6^2 - 5.6^2}$ in one go, press 2nd x^2 10.6 x^2 − 5.6 x^2) ENTER. The screen shows 9.

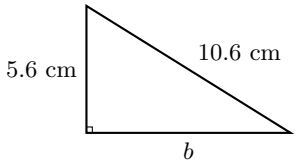
Example 1

Find the missing leg in each right triangle.

(a)



(b)



Right triangles are not always labelled a, b and c, and the right angle is not always at the bottom. Find the right angle first: the side across from it is the hypotenuse.

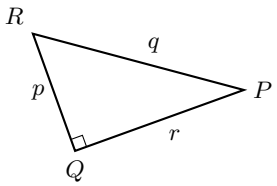
Example 2

(a) In $\triangle PQR$, the right angle is at Q. Each side is named by the small letter of the corner across from it.

(i) Which side is the hypotenuse? _____

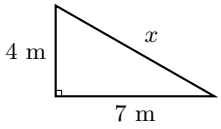
(ii) Write the Pythagorean theorem for $\triangle PQR$. _____

(iii) Rearrange it to find r. _____

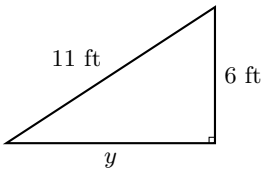


For (b) and (c), first decide: is the missing side a **leg** or the **hypotenuse**? Then find it to the nearest tenth.

(b)



(c)



2. Ladders, Kites and Other Real Problems

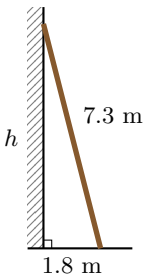
Many real problems hide a right triangle: a wall and the ground, a pole and its rope, a screen and its diagonal. **Draw a sketch**, mark the right angle, and label what you know. Then ask: is the missing side a leg or the hypotenuse?

Ladder safety. A common safety rule for a straight or extension ladder is the **4-to-1 rule**: for every 4 m of height up the wall, move the foot of the ladder 1 m out from the wall. So the distance out should be about the height $\div 4$. (This makes an angle of about 75° with the ground.)

Example 3

Jordan leans a 7.3 m extension ladder against the wall of a house in Red Deer. The foot of the ladder is 1.8 m from the wall.

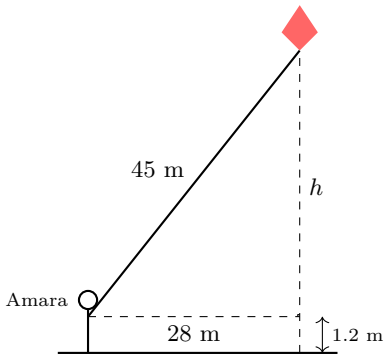
(a) How high up the wall does the ladder reach, to the nearest tenth of a metre?



(b) Does Jordan's ladder follow the 4-to-1 rule? Explain.

Example 4

Amara is flying a kite at Nose Hill Park. She holds the string 1.2 m above the ground. The string is 45 m long and pulled straight. The kite is directly above a spot that is 28 m from Amara.
How high is the kite above the ground, to the nearest tenth of a metre?



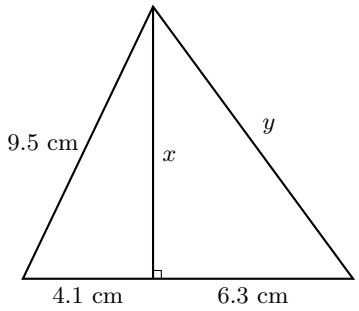
Watch out. Read the question again at the end. The triangle gave the height above Amara's *hand*, but the question asked for the height above the *ground*.

Some diagrams have **two** right triangles that share a side. Find the shared side from the first triangle, then use it in the second.

Keep the full value. Keep the full calculator value of the shared side (or its square) for the next step. Round only at the very end. Rounding in the middle can change your final answer.

Example 5

Find x and y, to the nearest tenth of a centimetre.



3. Is It a Right Triangle?

The Pythagorean theorem works **only** for right triangles. So we can use it as a test. Square the two shorter sides and add them. Compare the total with the square of the **longest** side.

Right triangle test. Let c be the **longest** side, and a and b the two shorter sides.

If $a^2 + b^2 = c^2$: the triangle **is** a right triangle. The right angle is across from c.

If $a^2 + b^2 > c^2$: not a right triangle. The biggest angle is **less** than 90° (the corner is too closed).

If $a^2 + b^2 < c^2$: not a right triangle. The biggest angle is **more** than 90° (the corner is too open).

Watch out. The longest side must be c, even when it is not written last. For sides 17, 8 and 15, test $8^2 + 15^2$ against 17^2.

Example 6

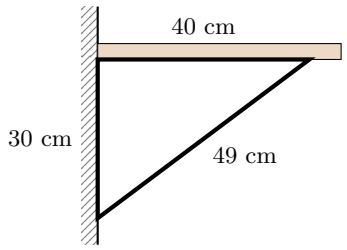
Is each triangle a right triangle? Complete the table.

Sides	**Shorter sides:** $a^2 + b^2$	**Longest side:** c^2	**Equal?**	**Right triangle?**
(a) 9 cm, 40 cm, 41 cm				
(b) 7 m, 9 m, 12 m				
(c) 3.3 ft, 5.6 ft, 6.5 ft				
(d) 15 in, 9 in, 13 in				

Example 7

In shop class, Harjit welds a steel shelf bracket. The part on the wall is 30 cm long, the part under the shelf is 40 cm long, and the brace between their ends is 49 cm long.

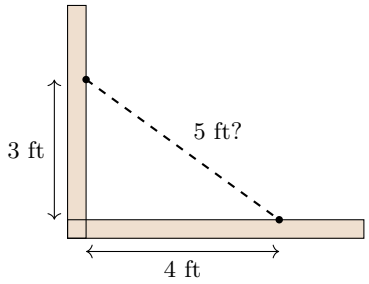
(a) Is the corner between the wall and the shelf a right angle? Show how you know.



(b) How long should the brace be to make a right angle?

4. Squaring a Corner: the 3-4-5 Method

A **square corner** is a 90° corner. Builders need square corners for decks, sheds, foundations, fences and garden beds. A carpenter's square is too small for a big job, but a tape measure and the Pythagorean theorem work for any size.



The 3-4-5 method

Step 1. Measure 3 units along one side from the corner and make a mark.

Step 2. Measure 4 units along the other side and make a mark.

Step 3. Measure the diagonal between the two marks. If it is exactly 5 units, the corner is square, because $3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

If the diagonal is **too long**, the corner is more than 90°: push the sides closer together. If it is **too short**, the corner is less than 90°: open the sides up.

Multiples of 3-4-5 also work: multiply all three numbers by the same number.

×2: 6-8-10 ×3: 9-12-15 ×4: 12-16-20 ×0.5: 1.5-2-2.5 (any unit: ft, m, . . .)

Use the **biggest** triangle that fits the job: a small measuring error matters less on a big triangle.

Builders have used this idea for thousands of years. Today framers, landscapers and concrete crews still use it, along with laser tools.

Example 8

Mateo is building a deck frame in his backyard in Airdrie. He measures 6 ft along one rim board from the corner and 8 ft along the other, and marks both spots. The diagonal between the marks is 10 ft 2 in.

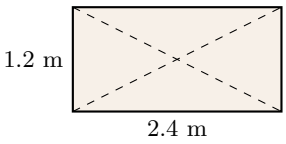
(a) What should the diagonal be? Why?

(b) Is the corner square? What should Mateo do?

Checking a rectangle with its diagonals. A rectangle has two diagonals (corner to opposite corner), and in a rectangle they are **equal**. If one diagonal is longer than the other, the corners are not square: the shape has been pushed over.

Example 9

Noor builds a frame for a raised garden bed. It should be a rectangle 1.2 m wide and 2.4 m long.
(a) How long should each diagonal be, to the nearest centimetre?



(b) Noor measures the two diagonals: 2.71 m and 2.65 m. Is her frame a rectangle with square corners? What should she do?

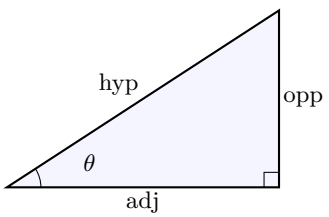
Lesson 4: Similar Right Triangles and the Tangent Ratio

Right triangles with the same acute angle are similar, so opposite ÷ adjacent is always the same number for that angle: its tangent. Use it to find a side you cannot measure.

1. Naming the Sides: Opposite, Adjacent and Hypotenuse

In a right triangle, the side across from the right angle is the **hypotenuse** (**hyp**). It is always the longest side. The other two sides are the **legs**.

Trigonometry means "triangle measurement". It uses one **acute** angle (less than 90°) of a right triangle to find lengths you can't measure directly. We often call that angle θ. It is the Greek letter **theta**: just a name for an angle, the way x is a name for a number. The two legs get names that tell where they are **from θ**.



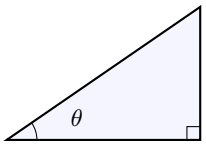
Labelling routine (do it first, every time)

Step 1. Find the right angle. The side across from it is the **hyp**.
Step 2. Find θ. The side across from θ is the **opp** (**opposite** side). It does not touch θ.
Step 3. The last side is the **adj** (**adjacent** side). It touches θ and it is not the hyp. (Adjacent means "next to".)

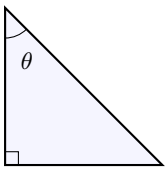
Watch out. The hyp touches θ too, but it is never called adj. And opp and adj depend on the angle you use: if you switch to the other acute angle, opp and adj **swap**. The hyp never changes.

Example 1

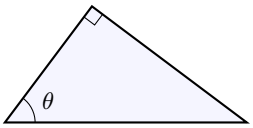
Write **opp**, **adj** or **hyp** beside each side. Use the angle marked θ.



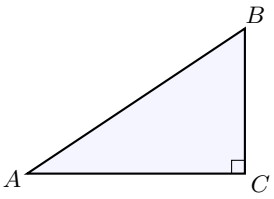
(a)



(b)



(c)



(d) In $\triangle ABC$, $\angle C = 90°$. Complete the table.

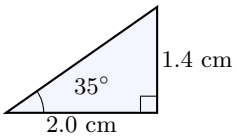
Side	**From $\angle A$, it is the ...**	**From $\angle B$, it is the ...**
BC		
AC		
AB		

2. Similar Right Triangles: opp ÷ adj Stays the Same

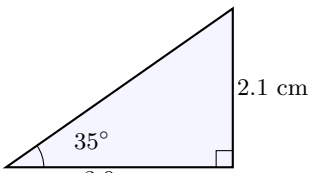
In Unit 6 you learned that two triangles are **similar** if two pairs of angles are equal. So all right triangles with the same acute angle are similar: they share the 90° angle and θ, and the third angle is $180° - 90° - \theta$ in each one. Similar triangles have the same shape. Let's see what that does to opp and adj.

Example 2

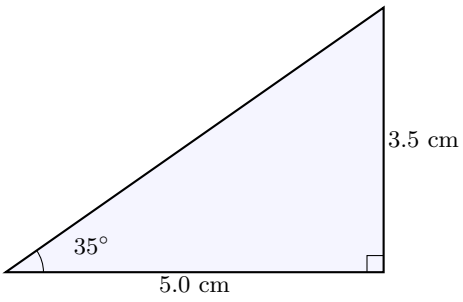
Three right triangles each have a 35° angle. They are drawn to actual size, so you can measure them with a ruler to check.



Triangle 1



Triangle 2



Triangle 3

(a) Why are the three triangles similar?

(b) Label opp and adj from the 35° angle. Then complete the table. Round opp ÷ adj to 2 decimal places.

	Triangle 1	**Triangle 2**	**Triangle 3**
opp (cm)			
adj (cm)			
opp ÷ adj			

(c) For *every* right triangle with a 35° angle, opp ÷ adj is about _____. The size of the triangle _____.

(d) A ramp has a 35° angle with the ground and is 10 m long along the ground (adj). Predict its height (opp).

Why it works. Triangle 2 is Triangle 1 enlarged by the scale factor $k = 1.5$, so *both* legs are multiplied by 1.5: $1.4 \times 1.5 = 2.1$ and $2.0 \times 1.5 = 3.0$. Multiplying the top and the bottom of a ratio by the same number does not change it. Each angle has its own value of opp ÷ adj, and it has a name.

The tangent ratio. In a right triangle, for an acute angle θ:

$$\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{\text{length of the side opposite } \theta}{\text{length of the side adjacent to } \theta}$$

For one angle, $\tan\theta$ is always the same number, however big or small the triangle is (the triangles are similar). We say "tan θ"; tan is short for **tangent**. The hyp is not used.

3. The tan Key on Your Calculator

You don't have to draw and measure a triangle to find $\tan\theta$. Your calculator knows it for every angle.

Calculator. Angles in this unit are in degrees, so the calculator must be in **DEG** mode: look for DEG on the screen. If it shows RAD or GRAD, press [DRG], use the arrow keys to underline DEG, and press [ENTER]. To find $\tan 35°$, press [tan] 35 [)] [ENTER]. The screen shows 0.700207538. Quick test of the mode: $\tan 45°$ must be exactly 1.

Example 3

(a) Use your calculator. Round each value to 4 decimal places.

θ	10°	20°	35°	45°	60°	70°	80°
$\tan\theta$							

(b) Does $\tan 35°$ match what you found in Example 2? _____

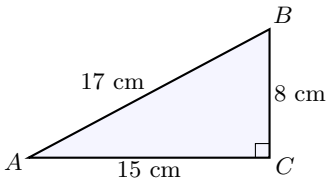
(c) As θ gets bigger, $\tan\theta$ gets _____. A steeper angle means the opp side is longer compared with the adj.

(d) $\tan 45° = 1$, so at 45° opp and adj are _____. For angles less than 45°, opp is _____ than adj. For angles more than 45°, opp is _____ than adj.

Example 4

In $\triangle ABC$, $\angle C = 90°$. Write each ratio as a fraction and as a decimal (to 4 decimal places).

(a) $\tan A$ (b) $\tan B$



(c) Which angle is bigger, $\angle A$ or $\angle B$? How do you know?

4. Finding a Missing Side: Multiply or Divide

If you know an acute angle and one leg, the tangent ratio gives you the other leg. Write the ratio, put in what you know, and solve for the unknown.

Finding a side with tan

Step 1. Label opp, adj and hyp from θ. (You need opp and adj: one you know, one you want.)

Step 2. Write $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$. **Step 3.** Substitute the numbers you know.

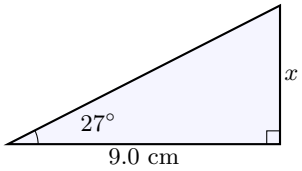
Step 4. Solve. x on the **top**: **multiply** both sides by the bottom number, so $x = \text{adj} \times \tan\theta$.
x on the **bottom**: multiply both sides by x, then **divide** both sides by $\tan\theta$, so $x = \text{opp} \div \tan\theta$.

Step 5. Round at the end, then check that the answer makes sense.

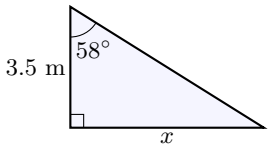
Calculator. For $9.0 \times \tan 27°$, press 9.0 [×] [tan] 27 [)] [ENTER]. For $6.3 \div \tan 40°$, press 6.3 [÷] [tan] 40 [)] [ENTER]. The [tan] key opens a bracket, so close it with [)] before you type anything after the angle. Don't round $\tan\theta$ on the way: type it all in one go.

Example 5

Find x, to the nearest tenth.



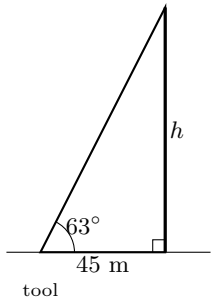
(a)



(b)

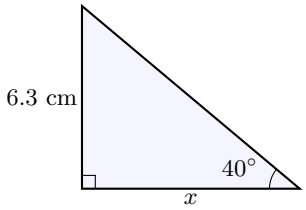
Example 6

A surveyor sets her angle tool on the ground, 45 m from the base of a cell-phone tower. The tool measures a 63° angle between the ground and the line to the top of the tower. How tall is the tower, to the nearest tenth of a metre?

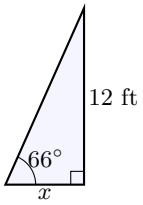


Example 7

Find x, to the nearest tenth. This time x is on the **bottom** of the ratio.



(a)



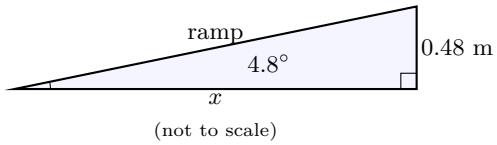
(b)

Watch out. $\tan 40°$ is a number ($0.8391\ldots$), not 40. Never write $x = 6.3 \div 40$. And when x is on the bottom, the answer is *not* $6.3 \times \tan 40°$: that would be the answer if x were the opp side.

Example 8

A wheelchair ramp must rise 0.48 m to reach a doorway. Accessibility guides suggest a ramp angle of about 4.8° or less.

(a) How far along the ground must the ramp reach? Round to the nearest tenth of a metre.



(not to scale)

(b) Many guides say a ramp should go 12 units along for every 1 unit up (a slope of 1 : 12). Does your answer to (a) agree?

5. Is the Answer Reasonable?

A calculator does what you type, even when it is wrong. Use these checks before you write your final answer.

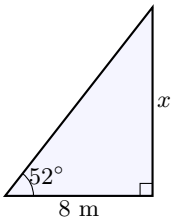
Quick checks for tan

- A length is always **positive**. A negative answer usually means the calculator is in RAD mode.
- **The 45° check:** if $\theta < 45°$, opp is shorter than adj. If $\theta > 45°$, opp is longer than adj. If $\theta = 45°$, they are equal.
- The hyp is the longest side, so each leg must be shorter than the hyp.
- Check with the other acute angle, $90° - \theta$: opp and adj swap, and you should get the same answer.

Example 9

A right triangle has a 52° angle, and the side adjacent to it is 8 m. Three students found the opposite side, x.

Ava	$x = 8 \div \tan 52° = 6.3$ m
Ben	$x = 8 \times \tan 52° = -48.4$ m
Cole	$x = 8 \times \tan 52° = 10.2$ m



(a) Who is correct? Use a check from the box above to show that the other two answers are not reasonable.

(b) What did Ava and Ben each do wrong?

Lesson 5: The Sine Ratio

For a given angle, the opposite side divided by the hypotenuse is always the same number. Use it to find the side across from an angle, or the hypotenuse.

1. Opposite ÷ Hypotenuse Stays the Same

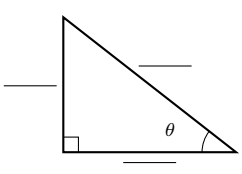
In Lesson 4 you named the sides of a right triangle from the angle you are using, θ ("theta").

Quick refresher. Labelling routine (Lesson 4). **Step 1.** Find the right angle. The side across from it is the **hyp**. It is always the longest side. **Step 2.** Find θ. The side across from θ is the **opp**. It does not touch θ. **Step 3.** The last side is the **adj**. It touches θ and it is not the hyp. If you switch to the other acute angle, opp and adj swap. The hyp never changes.

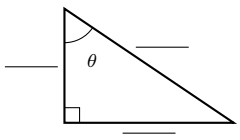
In Lesson 4 you divided opp by adj and got the **tangent**. In this lesson you divide **opp by hyp**.

Example 1

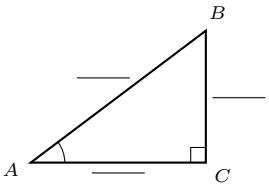
Label each side opp, adj or hyp for the marked angle.



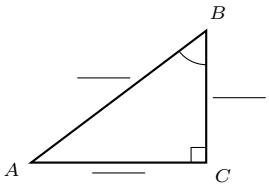
(a)



(b)



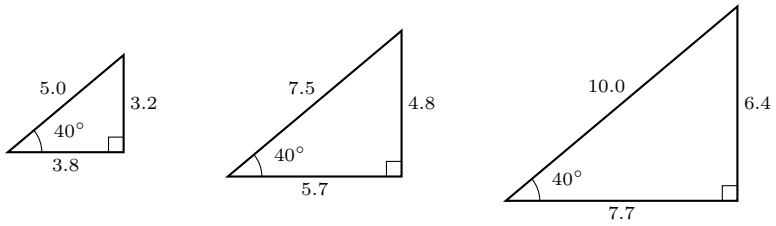
(c) For $\angle A$



(d) For $\angle B$

Example 2

Raj drew three right triangles. Each one has a 40° angle, so they are **similar** (Unit 6): same shape, different sizes. He measured every side to the nearest tenth of a centimetre.



Lengths in cm. The drawings are smaller than real size.

(a) Fill in the table. Round the last column to 2 decimal places.

Triangle	opp	hyp	opp ÷ hyp
Small			
Medium			
Large			

(b) What do you notice?

(c) Finish the sentence. In *every* right triangle with a 40° angle, the opposite side is about _____ of the hypotenuse.

The size of the triangle does not matter. Only the **angle** matters. This ratio has a name: the **sine** of the angle (say "sign"). We write $\sin 40°$.

The sine ratio. In a right triangle, for an acute angle θ:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{\text{length of the side opposite } \theta}{\text{length of the hypotenuse}}$$

Memory aid: **SOH** = **S**ine is **O**pposite over **H**ypotenuse.

2. Finding the Sine of an Angle

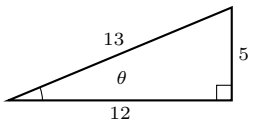
There are two ways to find the sine of an angle.

- From a triangle: divide the opposite side by the hypotenuse.
- From the angle: use the [sin] key on your calculator.

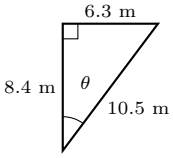
Calculator. Your TI-30XIIS must be in **DEG** (degree) mode: look for DEG on the screen. If it shows RAD or GRAD, press [DRG], use the arrow keys to underline DEG, and press [ENTER]. To find $\sin 40°$: [sin] 40 [)] [ENTER] shows 0.642787.... The [sin] key opens a bracket, so close it with [)]. Round sine values to **4 decimal places**: $\sin 40° \approx 0.6428$. That matches Raj's 0.64.

Example 3

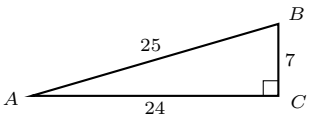
Write $\sin\theta$ as a fraction. Then write it as a decimal to 4 decimal places.



(a)



(b)



(c) Find $\sin A$ and $\sin B$.

Watch out. The hypotenuse always goes on the bottom. It is the longest side, so $\sin\theta$ is always **less than 1** for an acute angle. If you get a sine bigger than 1, you used the wrong sides.

Example 4

(a) Use your calculator. Round to 4 decimal places.

θ	$\sin\theta$	θ	$\sin\theta$
5°		45°	
20°		70°	
30°		89°	

(b) As the angle gets bigger, the sine gets _____.

(c) $\sin 30° = 0.5$. So with a 30° angle, the opposite side is _____ as long as the hypotenuse.

(d) Find $\sin 90°$. _____ Why does this make sense?

Check. Use $\sin 30° = 0.5$ to check answers in your head. For an angle a bit less than 30°, the opposite side is a bit less than half the hypotenuse. For a bigger angle, it is more than half.

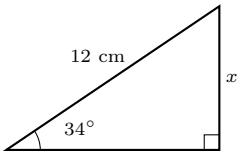
3. Finding the Opposite Side

If you know the hypotenuse and an angle, you can find the side across from the angle. The unknown is on **top** of the fraction, so you **multiply**.

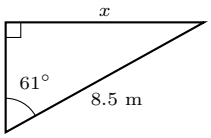
Finding a side with the sine ratio
Step 1. Label opp and hyp from the angle. Mark the side you know and the side you want (x).
Step 2. Write $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$ and substitute the numbers.
Step 3. Solve. Unknown on top (opp): **multiply** both sides by the hyp. Unknown on the bottom (hyp): **divide** (Section 4).
Step 4. Round at the end. Check: the hyp must be the longest side. Write the units (or a sentence).

Example 5

Find x to the nearest tenth.



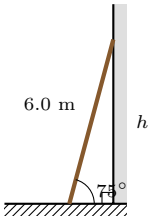
(a)



(b)

Example 6

Ladder labels say to set a ladder at about 75° to the ground. Kai has a 6.0 m ladder. He needs to reach a gutter that is 5.5 m above the ground. If he sets the ladder at 75°, will it reach the gutter?



4. Finding the Hypotenuse

If you know the opposite side and the angle, you can find the hypotenuse. Now the unknown is on the **bottom** of the fraction. Multiply both sides by the unknown to bring it up, then **divide** by the sine.

$$\sin 27° = \frac{4.5}{h} \quad \longrightarrow \quad h \times \sin 27° = 4.5 \quad \longrightarrow \quad h = \frac{4.5}{\sin 27°}$$

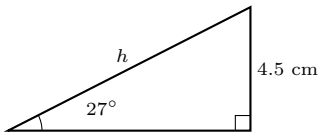
Unknown on top: $\text{opp} = \text{hyp} \times \sin\theta$ (multiply). **Unknown on the bottom:** $\text{hyp} = \dfrac{\text{opp}}{\sin\theta}$ (divide).

Calculator. For $h = 4.5 \div \sin 27°$, press 4.5 [÷] [sin] 27 [)] [ENTER]. It shows 9.912....

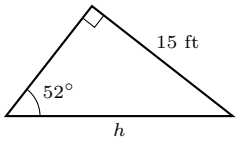
Example 7

Find the hypotenuse h to the nearest tenth.

(a)



(b)

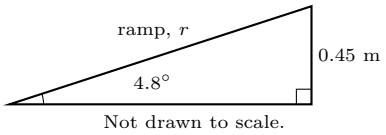


Example 8

A community hall in Okotoks needs a wheelchair ramp up to a door that is 0.45 m above the ground. Accessibility guides suggest a ramp angle of about 4.8° (a slope of about 1 : 12).

(a) How long must the ramp surface be? Round to the nearest tenth of a metre.

(b) Check: the 1 : 12 rule says about 12 m of ramp for each 1 m of rise. Does your answer agree?



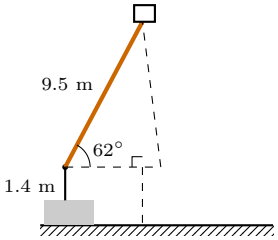
Not drawn to scale.

5. Solving Problems and Checking Your Answer

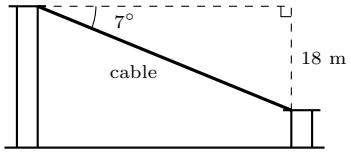
For a word problem, **sketch** a right triangle first. Put the angle in, label opp and hyp, and mark the side you want. Some problems need one more step at the end, such as adding a height.

Example 9

(a) A boom lift at a construction site has an arm 9.5 m long. The arm turns on a pivot 1.4 m above the ground. The arm is raised to 62° from the horizontal. How high above the ground is the worker's bucket at the end of the arm?



(b) A zipline at a summer camp starts on a tower and ends on a lower platform 18 m below the start. The cable makes a 7° angle with the horizontal. How long is the cable? (Treat it as straight.)



Not drawn to scale.

Watch out. Keep the full calculator value until the last step. In (a), add 1.4 to 8.388…, not to a rounded 8.4, and round only the final answer.

Example 10

Each student made one mistake. Explain how you can tell the answer is not reasonable, then fix it.

(a) A right triangle has opp $= 9$ and hyp $= 14$. Mia wrote $\sin\theta = \dfrac{14}{9} \approx 1.56$.

(b) A right triangle has opp $= 6$ m and a $35°$ angle. Theo wrote hyp $= 6 \times \sin 35° \approx 3.4$ m.

(c) Zoe found opp $= 10 \times \sin 50°$. Her calculator showed -2.6.

Lesson 6: The Cosine Ratio

In similar right triangles, the adjacent side divided by the hypotenuse never changes. Use this cosine ratio to find a side, and learn to choose between sine, cosine and tangent.

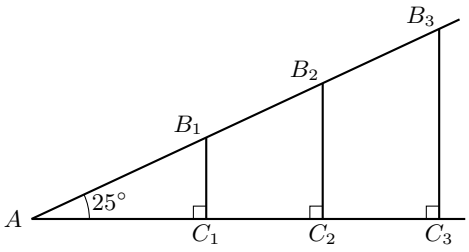
1. Adjacent ÷ Hypotenuse Stays the Same

In Lessons 4 and 5 you saw that **similar** right triangles (same shape, different size) keep the same ratio of sides. Opp ÷ adj gave the **tangent**, and opp ÷ hyp gave the **sine**. In this lesson you divide **adj by hyp**.

Quick refresher. hyp (hypotenuse): across from the right angle. It is always the longest side. **opp** (opposite): across from θ. It does not touch θ. **adj** (adjacent): next to θ, but not the hyp. If you change the angle, opp and adj swap. The hyp never changes.

Example 1

Three right triangles share the $25°$ angle at A, so they are similar. A student measured the sides.



Triangle	**adj** AC (cm)	**hyp** AB (cm)	**adj ÷ hyp** (2 decimals)
AB_1C_1	2.7	3	
AB_2C_2	4.5	5	
AB_3C_3	6.3	7	

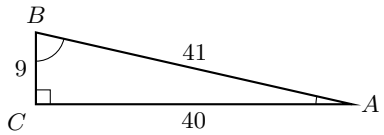
For a $25°$ angle, adj ÷ hyp is always about _____. The ratio does not depend on the _____ of the triangle. It depends only on the _____.

The cosine ratio. For an acute angle θ in a right triangle: $\cos\theta = \dfrac{\text{adj}}{\text{hyp}} = \dfrac{\text{length of the side adjacent to } \theta}{\text{length of the hypotenuse}}$

Say "cos" (short for **cosine**). Memory aid: **CAH** = **C**os is **A**dj over **H**yp.

Example 2

In $\triangle ABC$, $\angle C = 90°$. Complete the table for $\angle A$ and for $\angle B$.



	For $\angle A$	**For $\angle B$**
hyp		
adj		
opp		
cos as a fraction		
cos as a decimal (4 places)		

Watch out. opp and adj depend on **which angle** you use. Always mark the angle first, then label. The hypotenuse never changes.

2. Finding the Cosine of an Angle

You can find the cosine of an angle from a triangle (divide adj by hyp), or from the angle with the [cos] key on your calculator.

Calculator. Your TI-30XIIS must be in **DEG** (degree) mode: look for DEG on the screen. If it shows RAD or GRAD, press [DRG], move to DEG with the arrow keys, and press [ENTER]. To find $\cos 25°$: [cos] 25 [)] [ENTER] shows 0.906307787. Round cosine values to **4 decimal places**: $\cos 25° \approx 0.9063$. That is very close to the 0.90 we measured in Example 1. Quick mode test: $\cos 60°$ is exactly 0.5.

Example 3

(a) Find each value to 4 decimal places.

θ	10°	25°	40°	55°	70°	85°
$\cos\theta$						

(b) As the angle gets bigger, $\cos\theta$ gets _____.

(c) Every value is between _____ and _____.

(d) Why can $\cos\theta$ never be more than 1?

(e) Find $\cos 20°$ and $\sin 70°$. What do you notice? Use the triangle in Example 2 to explain.

Is my answer reasonable? $\cos\theta$ is always between 0 and 1. adj is always **shorter** than hyp. For a small angle, adj is almost as long as hyp. For an angle close to 90°, adj is very short.

3. Finding the Adjacent Side

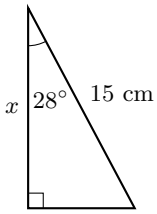
Use cosine when the question is about the **adj** and the **hyp**: you know one of them and you want the other one. If you want the adj, the unknown is on **top** of the fraction, so you **multiply**.

Finding a side with the cosine ratio
Step 1: Label adj and hyp from the angle. Mark the side you know and the side you want (x).
Step 2: Write $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$ and substitute the numbers.
Step 3: Solve. Unknown on top (adj): **multiply** both sides by the hyp. Unknown on the bottom (hyp): **divide** (Section 4).
Step 4: Round at the end. Check: the adj must be shorter than the hyp. Write the units (or a sentence).

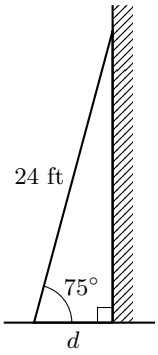
Calculator. For $15 \times \cos 28°$ press 15 [×] [cos] 28 [)] [ENTER]. Keep all the digits until the last step.

Example 4

(a) Find x to the nearest tenth of a centimetre.



(b) A 24-foot ladder leans against a wall. It makes a 75° angle with the ground (a safe angle). How far from the wall is the bottom of the ladder? Round to the nearest tenth of a foot.



4. Finding the Hypotenuse

If the unknown side is the hyp, it is on the **bottom** of the fraction. Multiply both sides by the unknown to get it off the bottom, then divide by $\cos\theta$.

$$\cos 41° = \frac{8.5}{x} \quad \longrightarrow \quad x \times \cos 41° = 8.5 \quad \longrightarrow \quad x = \frac{8.5}{\cos 41°}$$

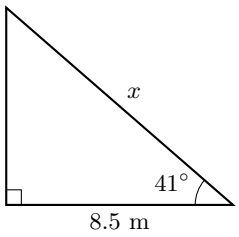
Unknown on top: $\text{adj} = \text{hyp} \times \cos\theta$ (multiply). **Unknown on the bottom:** $\text{hyp} = \dfrac{\text{adj}}{\cos\theta}$ (divide).

Calculator. For $x = 8.5 \div \cos 41°$, press 8.5 [÷] [cos] 41 [)] [ENTER]. It shows 11.262....

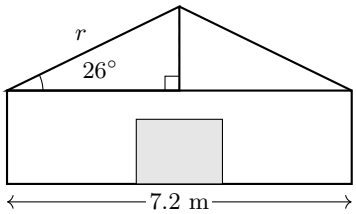
Watch out. If you multiply by mistake, you get a hyp that is *shorter* than the adj. That is impossible: the hyp is the longest side. Check every answer this way.

Example 5

(a) Find x to the nearest tenth of a metre.



(b) A garage is 7.2 m wide. Its roof rises from each wall to the middle at 26°. How long is each rafter r (from the top of the wall to the peak)? Round to the nearest tenth of a metre.



5. Sine, Cosine or Tangent? Choosing the Ratio

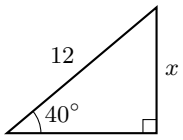
You now know three ratios. To choose, look at the two sides in the question: the side you **know** and the side you **want**.

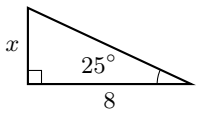
SOH CAH TOA $\sin\theta = \frac{\text{opp}}{\text{hyp}}$ (SOH) $\cos\theta = \frac{\text{adj}}{\text{hyp}}$ (CAH) $\tan\theta = \frac{\text{opp}}{\text{adj}}$ (TOA)

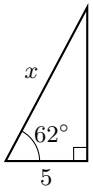
Label the sides. Circle the side you know and the side you want. Use the ratio that has **both** of them.

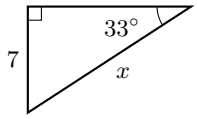
Example 6

For each triangle, name the side you know and the side you want (x). Then choose sin, cos or tan. Do not solve.









(a) Know: _____
Want: _____
Use: _____

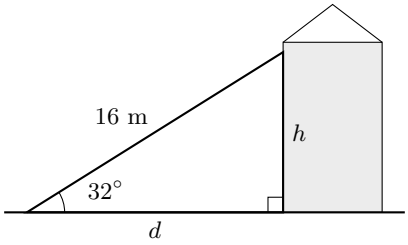
(b) Know: _____
Want: _____
Use: _____

(c) Know: _____
Want: _____
Use: _____

(d) Know: _____
Want: _____
Use: _____

Example 7

A grain auger is 16 m long. It leans against a grain bin and makes a 32° angle with the ground.



(a) How far from the bin is the bottom of the auger? Round to the nearest tenth of a metre.
(b) How high up the bin does the auger reach? Round to the nearest tenth of a metre.
(c) Check your answers with the Pythagorean theorem.

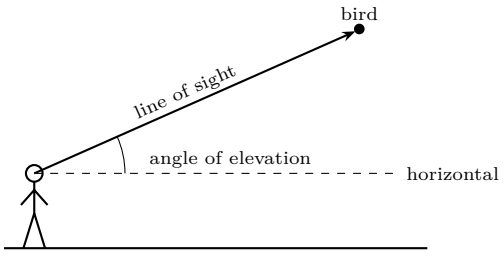
Lesson 7: Angles of Elevation and Depression

Measuring angles up or down from a level line, then using sin, cos and tan to find heights and distances you can't measure with a tape.

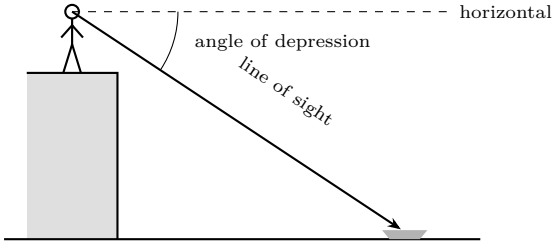
1. Looking Up and Looking Down

When you look at something, the straight line from your eye to the object is your **line of sight**. The **horizontal** is a level line (flat, like the floor or the surface of a calm lake) going out from your eye.

- The **angle of elevation** is the angle between the horizontal and your line of sight when you look **up**.
- The **angle of depression** is the angle between the horizontal and your line of sight when you look **down**.



Looking **up**



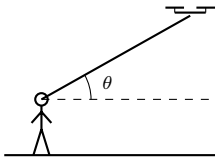
Looking **down**

Angle of elevation: from the horizontal **up** to the line of sight. **Angle of depression:** from the horizontal **down** to the line of sight.

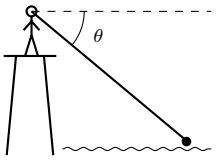
Both are always measured from the **horizontal** (level) line, never from a wall or a pole.

Example 1

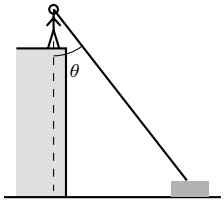
Is the marked angle θ an angle of elevation, an angle of depression, or neither? Explain.



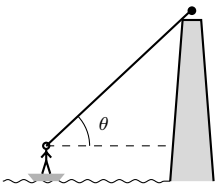
(a) a drone



(b) a lifeguard



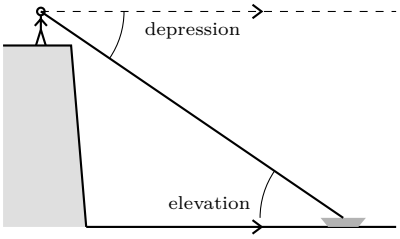
(c) a rooftop



(d) a lighthouse

2. The Angle of Elevation Equals the Angle of Depression

Suppose Jada stands on top of a cliff and looks down at a boat, while someone in the boat looks up at her. Both people use the **same line of sight**. The horizontal at the top and the horizontal at the bottom (the water) are **parallel**, and the line of sight is a **transversal**. So the two angles are **alternate interior angles** (Unit 5, the Z shape), and they are equal.



Angle of depression = angle of elevation
The angle of depression from the top equals the angle of elevation from the bottom. (They are alternate interior angles between two parallel horizontal lines.)

Watch out. If an angle is measured from the **vertical** (a wall, a cliff face, a pole), it is not the angle of elevation or depression. Subtract it from 90°.

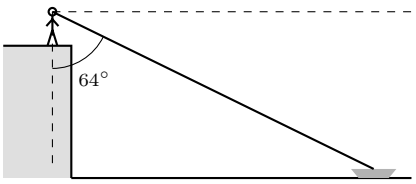
This is very useful: an angle of depression is at the *top*, outside the triangle, but the equal angle of elevation is at the *bottom*, inside the right triangle, where you can use it.

Example 2

(a) From the top of a grain elevator, Mia sees her truck at an angle of depression of 38°. What is the angle of elevation from the truck to Mia?

(b) A drone pilot looks up at her drone at an angle of elevation of 41°. What is the angle of depression from the drone to the pilot?

(c) Ken looks down from the top of a cliff at a canoe. His line of sight makes an angle of 64° with the cliff face (the vertical). Find the angle of depression.



(c)

3. Making and Reading a Clinometer

A **clinometer** is a tool that measures an angle of elevation or depression. Surveyors and foresters use electronic ones, but you can make one in a few minutes.

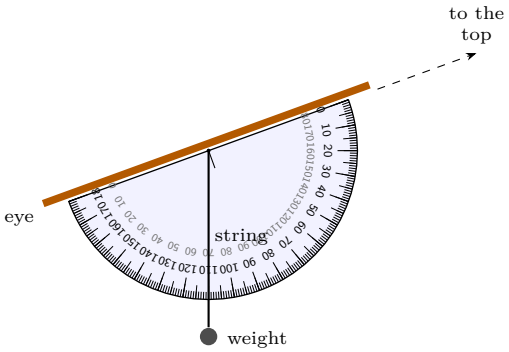
Make your own:

1. Tape a drinking straw along the straight edge of a protractor.

2. Tie a string through the small hole at the centre.

3. Tie a weight (a washer or a few paper clips) to the other end, so the string hangs below the curved edge.

Use it: Hold the protractor with the curved side down. Look through the straw at the top of the object. Let the string hang straight down, then pinch it against the protractor and read the number where it crosses the scale.



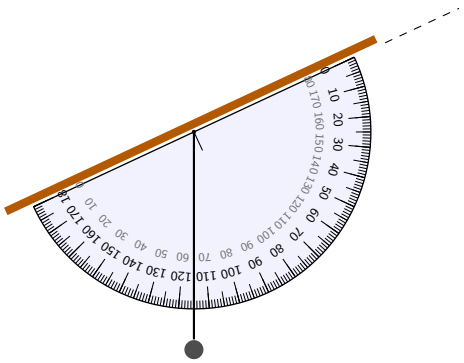
When the straw is level, the string crosses at 90°. Tilt the straw up by 25° and the string crosses at 65° on one scale and 115° on the other. Either way, it is 25° away from 90°.

Reading a clinometer: angle of elevation (or depression) = the difference between the reading and 90°
reading less than 90°: angle = 90° − reading reading more than 90°: angle = reading −90°

Example 3

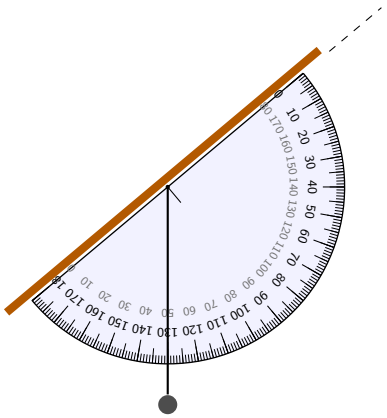
(a), (b) Read each clinometer. Then find the angle of elevation.

(a)



Reading: _____ Angle: _____

(b)



Reading: _____ Angle: _____

(c) Find the angle of elevation for each clinometer reading.

Reading	72°	58°	113°	90°	134°
Angle of elevation					

4. Finding a Height: Don't Forget Your Eyes

Now put the angle into a right triangle and use the trig ratios from Lessons 4 to 6.

Quick refresher. SOH CAH TOA: $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$. Label opp and adj from the angle you are using. If the unknown is on top, multiply. If it is on the bottom, divide.

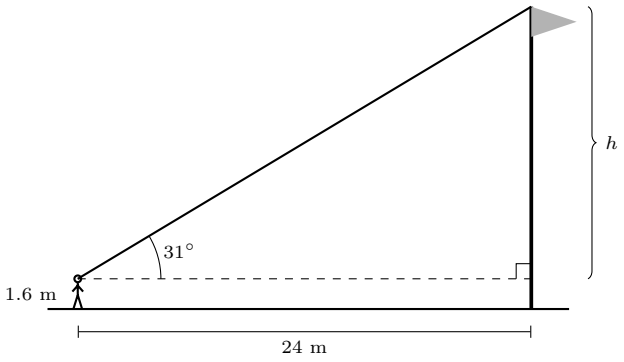
Solving an elevation or depression problem
Step 1: Sketch it. Draw the horizontal, the line of sight, and mark the angle.
Step 2: Find the **right triangle**. For an angle of depression, use the equal angle of elevation at the bottom.
Step 3: Label **opp, adj, hyp** from that angle. Mark the side you know and the side you want (x).
Step 4: Choose the ratio that uses those two sides. Write it, substitute, solve.
Step 5: If the angle was measured from your eyes (or hand), **add that height**. Check, then write a sentence.

An angle of elevation measured by a person starts at their **eyes**, not at the ground. The triangle only reaches up from eye level, so the tan or sin answer is the height *above your eyes*.

Calculator. Your TI-30XIIS must be in **DEG** (degree) mode: look for DEG on the screen. If it shows RAD or GRAD, press [DRG], move to DEG with the arrow keys, and press [ENTER]. To find $24 \times \tan 31°$, press 24 [×] [tan] 31 [)] [ENTER].

Example 4

Noor stands 24 m from the base of the flagpole at her school. Her eyes are 1.6 m above the ground. The angle of elevation to the top of the pole is 31°. How tall is the flagpole, to the nearest tenth of a metre?



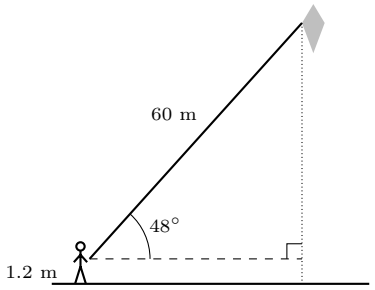
Check with 45°. At 45°, the height above your eyes **equals** your distance from the object (the triangle has two equal legs). If the angle is less than 45°, the height above your eyes is **less** than the distance. If the angle is more than 45°, it is **more**.

Example 5

Liam flies a kite on 60 m of string. He holds the string 1.2 m above the ground. The string makes an angle of elevation of 48°. Assume the string is straight. Round to the nearest tenth.

(a) How high is the kite above the ground?

(b) How far is Liam from the spot on the ground right under the kite?



Activity (optional): measure a real height with your clinometer. Work with a partner. Pick something tall at school (a wall, a light pole, a tree, the gym). Measure your distance from its base with a tape or by pacing, and measure your eye height. Sight the top through the straw, read the clinometer, and find the height.

Object	Distance to base	Eye height	Clinometer reading	Angle of elevation	Height above eyes	Height of object

5. Looking Down: Distances from Above

For an angle of depression, the angle is at the top, *outside* the triangle. Move it to the bottom: the angle of elevation there is equal. Then the height is opp, and the distance along the ground is adj.

Watch out. When the distance you want is adj and you know opp, $\tan\theta = \dfrac{\text{opp}}{x}$ puts x on the **bottom**, so you **divide**: $x = \dfrac{\text{opp}}{\tan\theta}$. On the calculator: 45 $\boxed{\div}$ $\boxed{\tan}$ 17 $\boxed{)}$ $\boxed{\text{ENTER}}$.

Example 6

From a lookout on a cliff, Ava is 45 m above a lake. She sees a canoe at an angle of depression of 17°. How far is the canoe from the base of the cliff, to the nearest tenth of a metre? Sketch it first.

Example 7

A plane is 1800 m above the runway when it starts its descent to the Calgary airport. It comes down in a straight line at an angle of depression of 4°. Round to the nearest tenth of a kilometre.

(a) What is the horizontal distance from the plane to the spot where it touches down?

(b) How far does the plane fly along its path to touch down?

Lesson 8: Finding Angles with Inverse Trig Ratios

When you know two sides of a right triangle, the inverse keys on your calculator turn their ratio back into the angle.

1. Undoing a Trig Ratio

In Lessons 4 to 7 you started with an *angle* and used [sin], [cos] or [tan] to get a *ratio*. For example, $\tan 35° \approx 0.7002$.

Now we go the other way: we know the ratio and want the angle. To go backwards, we use an **inverse** trig ratio. "Inverse" means "undo", the way subtracting undoes adding and the square root undoes squaring.

Inverse trig ratios (read $\sin^{-1}$ as "inverse sine")

If $\sin\theta = 0.5$,	then $\theta = \sin^{-1}(0.5) = 30°$.	"Which angle has a sine of 0.5?"
If $\cos\theta = 0.2$,	then $\theta = \cos^{-1}(0.2) \approx 78°$.	"Which angle has a cosine of 0.2?"
If $\tan\theta = 0.7002$,	then $\theta = \tan^{-1}(0.7002) \approx 35°$.	"Which angle has a tangent of 0.7002?"

Calculator. First check that the screen shows **DEG** (degree mode). If it shows RAD or GRAD, press [DRG], choose DEG with the arrow keys and press [ENTER].
The inverse keys are the **second functions** of the trig keys: [2nd][sin] gives $\sin^{-1}$, [2nd][cos] gives $\cos^{-1}$, and [2nd][tan] gives $\tan^{-1}$.
Example: [2nd][tan] 0.7002 [)] [ENTER]. The screen shows $\tan^{-1}(0.7002)$ and the answer 34.99…, so the angle is about 35°.

Watch out. The -1 is **not** a power. $\tan^{-1}(0.7002)$ does *not* mean $1 \div 0.7002$. It is the angle whose tangent is 0.7002. Round angles to the **nearest degree** unless the question says otherwise.

Example 1

Find each angle θ to the nearest degree.

	Write it with an inverse	Calculator shows	θ (nearest degree)
(a) $\sin\theta = 0.6157$			
(b) $\cos\theta = 0.3420$			
(c) $\tan\theta = 0.4663$			
(d) $\sin\theta = 0.95$			
(e) $\cos\theta = 0.875$			
(f) $\tan\theta = 2.5$			

When the ratio is a fraction. In a triangle the ratio usually starts as a fraction, like $\sin A = \frac{5}{8}$. Type the whole fraction inside the bracket. Do not round the fraction to a short decimal first: that can change the angle.

Calculator. $\sin A = \frac{5}{8}$: [2nd][sin] 5 [÷] 8 [)] [ENTER] shows 38.68…, so $\angle A \approx 39°$.

Example 2

(a) $\sin A = \dfrac{5}{8}$. Find $\angle A$.

(b) $\cos B = \dfrac{4}{11}$. Find $\angle B$.

(c) $\tan C = \dfrac{12}{7}$. Find $\angle C$.

(d) For part (b), Maya rounded $\frac{4}{11}$ to 0.4 first. What angle did she get? Why is it wrong?

2. Finding an Angle From Two Sides

To find an acute angle θ in a right triangle, you need **two sides**. Label them from θ's point of view, as in Lessons 4 to 6: the **hypotenuse** (hyp) is across from the right angle and is the longest side; the **opposite** side (opp) is across from θ; the **adjacent** side (adj) touches θ and is not the hypotenuse.

Finding an angle

Step 1: Label the sides opp, adj and hyp from the angle you want.

Step 2: Circle the two sides you know. Choose the ratio that uses them (SOH CAH TOA):

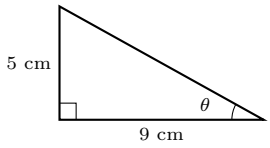
opp and hyp $\rightarrow \sin\theta = \dfrac{\text{opp}}{\text{hyp}}$ adj and hyp $\rightarrow \cos\theta = \dfrac{\text{adj}}{\text{hyp}}$ opp and adj $\rightarrow \tan\theta = \dfrac{\text{opp}}{\text{adj}}$

Step 3: Write the ratio and put in the two lengths.

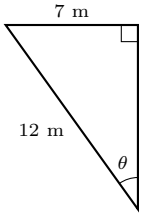
Step 4: Use the inverse key: $\theta = \sin^{-1}(\ldots)$, $\cos^{-1}(\ldots)$ or $\tan^{-1}(\ldots)$. Round to the nearest degree.

Example 3

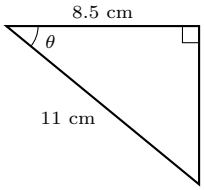
Label the two given sides opp, adj or hyp from θ. Then find θ to the nearest degree.



(a)



(b)



(c)

3. The Other Acute Angle, and Checking

The three angles of any triangle add to 180°. In a right triangle one angle is 90°, so the two **acute angles add to 90°**.

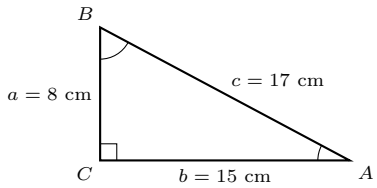
Other acute angle $= 90° - \theta$ (for example, if one acute angle is 28°, the other is $90° - 28° = 62°$)

You can **check** an angle by finding it a second way: with a different ratio, or from the other angle's point of view. Both ways should give the same answer.

Watch out. When you move to the other acute angle, **opp and adj swap**. The side that was opposite the first angle is adjacent to the second one. The hypotenuse never changes.

Example 4

In $\triangle ABC$, $\angle C = 90°$. The sides are $a = 8$ cm, $b = 15$ cm and $c = 17$ cm.



(a) Find $\angle A$ with the tangent ratio.

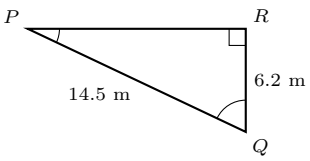
(b) Check $\angle A$ with the sine ratio.

(c) Find $\angle B$ by subtracting.

(d) Check $\angle B$ with a ratio from B's point of view.

Example 5

In $\triangle PQR$, $\angle R = 90°$, the hypotenuse $PQ = 14.5$ m and $QR = 6.2$ m. Find $\angle P$ and $\angle Q$ to the nearest degree. Check $\angle Q$ with a ratio.



4. Is the Angle Reasonable?

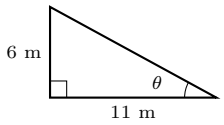
A calculator does exactly what you type, so a small slip gives a wrong angle. Before you write your answer, compare it with these benchmarks.

Quick checks for an angle θ in a right triangle

- If the two legs are **equal**, $\theta = 45°$. If opp is **shorter** than adj, θ is **less than** $45°$. If opp is **longer**, θ is **more than** $45°$.
- If opp is **half** the hypotenuse, $\theta = 30°$ ($\sin 30° = 0.5$). If adj is half the hypotenuse, $\theta = 60°$.
- The bigger acute angle is across from the longer leg.
- An **ERROR** with $\sin^{-1}$ or $\cos^{-1}$ means the number was more than 1. The hypotenuse is the longest side, so it always goes on the **bottom**.
- A tiny answer like 0.5 (for an angle that looks like about 30°) means the calculator is in RAD mode.

Example 6

A right triangle has legs of 6 m and 11 m. Ava, Ben and Cara each tried to find the angle θ across from the 6 m leg. Each one made a mistake. Find it and fix it.



Student's work	**What went wrong? (Use a quick check.)**	**Correct work**
(a) Ava: $\theta = \tan^{-1}\left(\frac{11}{6}\right) \approx 61°$		
(b) Ben: $\tan^{-1}\left(\frac{6}{11}\right)$ shows 0.4993		
(c) Cara (a different triangle: adj 9.6 m, hyp 15 m): $\cos^{-1}\left(\frac{15}{9.6}\right)$ gives DOMAIN ERROR		

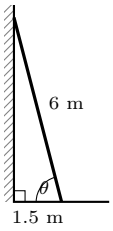
5. Angles on the Job: Ladders, Ramps and Roofs

Many jobs need an angle that you cannot measure easily, but you *can* measure two lengths. Sketch a right triangle, mark the right angle, write the lengths you know, mark the angle you want, then follow the four steps.

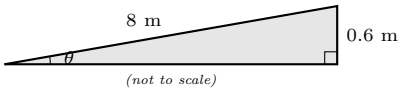
Safety guidelines (approximate; always follow the rules for your job site)
Ladders: a ladder should make an angle of **about 75°** with the ground. (Rule of thumb: 1 m out from the wall for every 4 m of ladder.)
Wheelchair ramps: a slope of 1 in 12 (1 m up for every 12 m along the ground), which is **about 4.8°**, is the steepest usually allowed.
Roof pitch: a "5/12 pitch" means the roof rises 5 inches for every 12 inches across.

Example 7

(a) A painter leans a 6 m ladder against a wall. The foot of the ladder is 1.5 m from the wall. What angle does the ladder make with the ground? Is it safe?

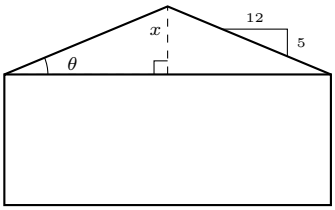


(b) A wheelchair ramp at a community hall is 8 m long (along the ramp surface). It rises 0.6 m. Find the angle between the ramp and the ground, to the nearest tenth of a degree. Does it meet the guideline?

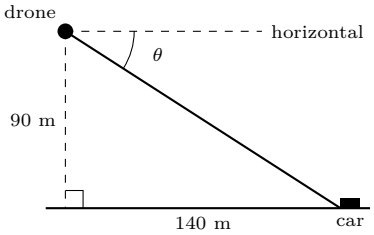


Example 8

(a) A garage roof has a 5/12 pitch. Find the angle θ between the roof and the horizontal. Then find angle x at the peak, between the rafter and the vertical centre post.



(b) A drone is hovering 90 m above a field. A car is parked 140 m away, measured along the ground. Find the angle of depression from the drone to the car.



Lesson 9: Solving Right Triangles: Choosing the Right Ratio

A four-step routine for picking Pythagoras, sine, cosine or tangent, finding every side and angle of a right triangle, working with two triangles at once, and checking that answers make sense.

1. Which Tool Do I Use? A Four-Step Routine

You now have four tools for right triangles: the **Pythagorean theorem**, the three **trigonometric ratios** (sine, cosine and tangent), their **inverse keys** for angles, and the fact that the two acute angles add to 90°. The hard part is often not the math. It is **choosing** the right tool. This routine works every time.

Solving a right triangle: the 4-step routine

1. Sketch and label. Mark the right angle and the angle you are using (θ). Label the sides **opp**, **adj** and **hyp** *from* θ. Put a ? on what you want.

2. Choose the tool. Look at what you **know** and what you **want**:

You know	You want	Use
two sides	the third side	Pythagoras: $a^2 + b^2 = c^2$
one acute angle	the other acute angle	$90° -$ the angle you know
an acute angle and one side	another side	the ratio with the side you know and the side you want
two sides	an acute angle	the ratio with those two sides, then 2nd and the ratio key

3. Solve. Write the equation, substitute, solve. Unknown on **top**: multiply. Unknown on the **bottom**: divide.

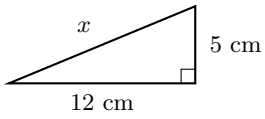
4. Check. Is the answer reasonable? (See Section 5.)

SOH CAH TOA: $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$ $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$ $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$

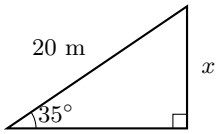
Leave one out. To choose a ratio, find the side you do **not** know and do **not** want. Leave it out. No hyp → TOA. No opp → CAH. No adj → SOH.

Example 1

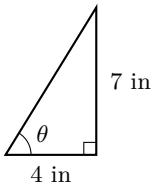
For each triangle, name the tool, write the equation, then solve. Round sides to the nearest tenth and angles to the nearest degree.

(a) 

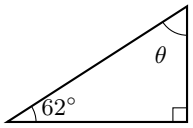
Tool: _____

(b) 

Tool: _____

(c) 

Tool: _____

(d) 

Tool: _____

Calculator. Check that **DEG** shows on the screen. If you see RAD or GRAD, press [DRG], choose DEG and press [ENTER]. (Test: $\sin 30°$ should give 0.5.)
Unknown on top: 20 [×] [sin] 35 [)] [ENTER]. Unknown on the bottom: 9 [÷] [sin] 31 [)] [ENTER].
An angle: [2nd] [tan] 7 [÷] 4 [)] [ENTER]. A square root: [2nd] [x^2] 5 [x^2] [+] 12 [x^2] [)] [ENTER].

2. Solving a Whole Triangle

To **solve a triangle** means to find *all* of its missing sides and angles. A right triangle has three sides and two acute angles (the right angle is already known). If you know **two sides**, or **one side and one acute angle**, you can find everything else.

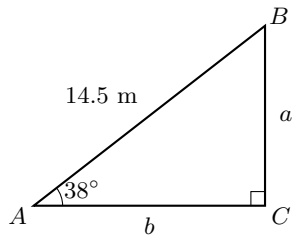
Tips for solving a whole triangle
- If you know an acute angle, find the other one first: it is just $90° -$ the angle.
- Each time, use the numbers you were **given** if you can, not an answer you rounded. Rounded numbers make small errors grow.
- Keep the full calculator value until the end. Round sides to the nearest tenth and angles to the nearest degree.
- Finish with a check: Pythagoras, or the angles adding to $90°$.

Example 2

Solve $\triangle ABC$. $\angle C = 90°$, $\angle A = 38°$ and the hypotenuse $c = 14.5$ m.

$\angle B =$ _____ $a =$ _____

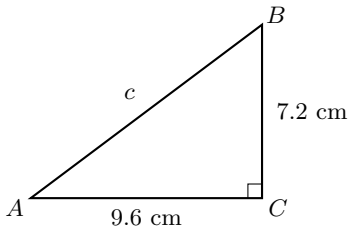
$b =$ _____ Check: _____



Example 3

Solve $\triangle ABC$. $\angle C = 90°$, and the two legs are $a = 7.2$ cm and $b = 9.6$ cm.

$c =$ _____ $\angle A =$ _____ $\angle B =$ _____

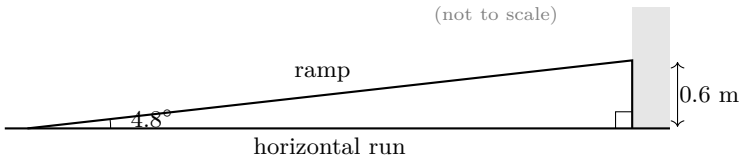


3. Solving Triangles in Real Problems

In a word problem, the triangle is hidden in the story. Draw it first. Ground and floors are **horizontal**; walls, poles and towers are **vertical**, so they meet at a right angle. Then use the same 4 steps.

Example 4

A community hall in Airdrie needs a wheelchair ramp up to a door that is 0.6 m above the sidewalk. Canadian accessibility guidance says a ramp should rise at an angle of about 4.8° or less (that is about 1 m up for every 12 m along).



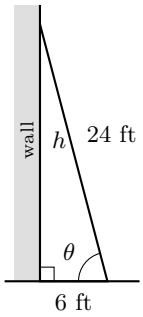
(a) How long must the ramp be?

(b) How much horizontal space along the sidewalk does the ramp need?

Watch out. When the unknown is on the **bottom**, students often multiply by mistake: $0.6 \times \sin 4.8° = 0.05$ m. A ramp 5 cm long can't climb 60 cm! The hypotenuse must be the *longest* side.

Example 5

A painter in Lethbridge leans a 24 ft extension ladder against a house. The foot of the ladder is 6 ft from the wall. A ladder is safest at about 75° to the ground.
(a) What angle does the ladder make with the ground? Is it safe?



(b) How high up the wall does the ladder reach?

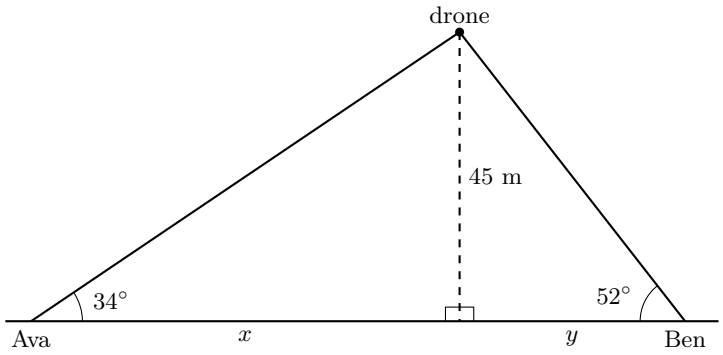
4. Two Right Triangles in One Problem

Some problems have **two** right triangles that share a side. The shared side is the key.

Two-triangle strategy
1. Sketch both triangles. Mark the side they **share**.
2. Start in the triangle where you know **enough** (two facts besides the right angle).
3. Use what you found in the first triangle (keep the full calculator value) to solve the second triangle.
4. Add or subtract lengths if the question asks for a total or a difference.

Example 6

A drone is hovering 45 m above a soccer field near Red Deer. Ava and Ben stand on opposite sides of the point right under the drone. Ava sees the drone at an angle of elevation of 34°. Ben sees it at 52°. How far apart are Ava and Ben?



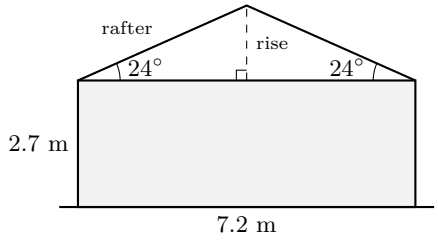
Example 7

A garage is 7.2 m wide and its walls are 2.7 m high. The two sides of the roof meet in the middle, and each side rises at 24°.

(a) How high does the roof rise above the walls?

(b) How long is each rafter (from wall to peak)?

(c) How tall is the garage at its peak?



5. Is the Answer Reasonable?

A calculator does exactly what you type, even when it is wrong. Always ask: *does this answer make sense?*

Reasonableness checks for right triangles

- The **hypotenuse is the longest side**. Each leg is shorter than it.
- The two acute angles **add to 90°**.
- The **bigger angle faces the longer side**; the smaller angle faces the shorter side.
- $\sin\theta$ and $\cos\theta$ are always **less than 1** (the hyp is on the bottom). If the inverse key gives an ERROR, you probably put the hyp on top.
- **Benchmarks:** at 45° the two legs are equal; at 30° the opposite side is half the hypotenuse.
- Check the calculator is in **DEG** mode ($\sin 30° = 0.5$).

Example 8

Each answer below is wrong. Explain how you can tell without solving, then find the correct answer.

(a) The legs of a right triangle are 9 m and 4 m. Jordan says the hypotenuse is 8.5 m.

(b) In a right triangle, the side opposite θ is 8 cm and the hypotenuse is 12 cm. Priya pressed [2nd] [sin] 12 [÷] 8 [)] [ENTER] and got an ERROR.

(c) In a right triangle with a 20° angle and a 70° angle, Leo says the leg opposite the 20° angle is 15 cm and the leg opposite the 70° angle is 5.5 cm.

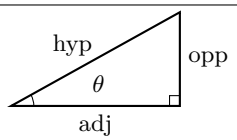
(d) Sam stands 20 m from a flagpole and measures a 35° angle of elevation from the ground to the top. He gets a height of $20\tan 35° = 9.5$ m.

Lesson 10: Unit 7 Review

Every rule from the unit on one page, worked examples that mix the topics, and a practice unit test.

1. The Unit on One Page

Use this page to study. Each row is one skill: what to do, and a short example. Lengths are rounded to the nearest tenth and angles to the nearest degree unless a question says otherwise.

Skill	What to do	Example
Squares and square roots	Squaring multiplies a number by itself. The **square root** undoes it ([2nd] [x^2] is $\sqrt{\ }$). Estimate a root between the two perfect squares around it.	$9^2 = 81$, $\sqrt{81} = 9$ $\sqrt{50}$ is between 7 and 8 (49 and 64): ≈ 7.1
Triangle basics	The angles of any triangle add to 180°. A **right triangle** has a 90° angle (square mark). The two sides that make the right angle are the **legs**; the side across from it is the **hypotenuse**, the longest side.	Angles 90° and 35°: third angle $= 180° - 90° - 35° = 55°$
Pythagorean theorem	Right triangles only: $a^2 + b^2 = c^2$ (c is the hypotenuse). Hypotenuse: $c = \sqrt{a^2 + b^2}$ (add). A leg: $a = \sqrt{c^2 - b^2}$ (subtract).	Legs 5 and 12: $c = \sqrt{25 + 144} = \sqrt{169} = 13$
Right triangle test; square corners	Is $a^2 + b^2$ equal to c^2 (with c the longest side)? Equal: right triangle. Not equal: not a right triangle. **3-4-5 method**: measure 3 along one side and 4 along the other; the diagonal is 5 if the corner is square (also 6-8-10, 9-12-15). A rectangle is square if its two diagonals are equal.	6, 7, 9: $36 + 49 = 85$, $9^2 = 81$; $85 \neq 81$, so not a right triangle
Naming the sides	Stand at the angle you are using, θ. **hyp**: across from the right angle. **opp**: across from θ. **adj**: next to θ (not the hyp). Change the angle and opp and adj swap.	
Trig ratios	**SOH CAH TOA**: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$. Similar right triangles with the same θ have the same ratios. Calculator in **DEG** mode.	$\tan 50° \approx 1.1918$: in every right triangle with a 50° angle, opp is about 1.19 times adj
Finding a side	Label the sides, pick the ratio that uses the side you know and the side you want, substitute, solve. Unknown on **top**: multiply. Unknown on the **bottom**: divide.	$\tan 40° = \frac{x}{10}$: $x = 10 \times \tan 40° \approx 8.4$ $\sin 40° = \frac{6}{x}$: $x = 6 \div \sin 40° \approx 9.3$
Finding an angle	Use two sides and the inverse key: $\theta = \tan^{-1}\left(\frac{\text{opp}}{\text{adj}}\right)$ with [2nd] [tan] (also $\sin^{-1}$, $\cos^{-1}$). The other acute angle is $90° - \theta$.	opp 4, adj 7: $\theta = \tan^{-1}(4 \div 7) \approx 30°$; other angle 60°
Elevation and depression	Both are measured from the **horizontal**. Elevation: looking up. Depression: looking down. The angle of depression from the top equals the angle of elevation from the bottom (alternate interior angles). Add the eye (or hand) height when you sight from above the ground.	Eye 1.5 m, 20 m away, elevation 50°: $20\tan 50° + 1.5 \approx 25.3$ m
Solving a right triangle	Find every missing side and angle. Routine: sketch and label, choose Pythagoras or SOH CAH TOA, solve, check. Two triangles that share a side: solve the one with enough facts first.	Checks: hyp is longest; the bigger angle faces the longer side

Remember. Calculator in **DEG** mode (the screen shows DEG; if not, press [DRG] and choose DEG). Label opp, adj and hyp from the angle you are using. Keep the full calculator value until the last step, then round. Check: the hypotenuse is the longest side, and $\sin\theta$ and $\cos\theta$ are always less than 1.

2. Square Roots and the Pythagorean Theorem

Example 1

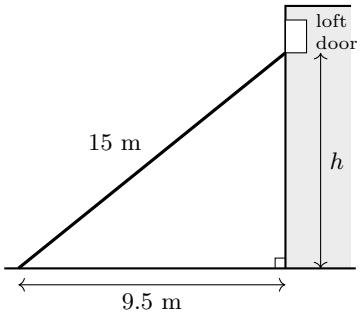
(a) $\sqrt{70}$ is between which two whole numbers? Estimate it, then find it to the nearest tenth.

(b) A triangle has angles of 90° and 34°. Find the third angle. What kind of triangle is it?

(c) A right triangle has legs of 9 cm and 14 cm. Find the hypotenuse.

Example 2

A hay bale elevator (a long conveyor) is 15 m long. Its lower end sits on the ground, 9.5 m from the wall of a barn. Its top end rests at the bottom of the hayloft door. How high is the hayloft door, to the nearest tenth of a metre?



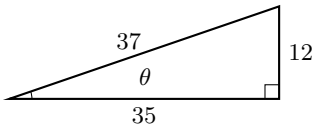
Example 3

(a) Gita is framing a shed floor 2.4 m by 3.2 m. What should the diagonal be if the corners are square? She measures 4.06 m. Are the corners square?

(b) Is a triangle with sides 16 cm, 63 cm and 65 cm a right triangle? Explain using the Pythagorean theorem.

3. The Trig Ratios: Finding Sides

Example 4

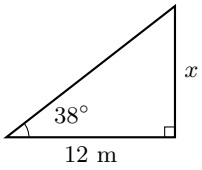


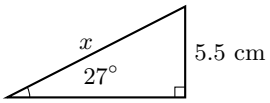
(a) Label opp, adj and hyp for θ. Then write $\sin\theta$, $\cos\theta$ and $\tan\theta$ as fractions and as decimals (to 4 decimal places).

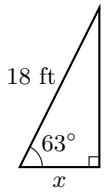
(b) A bigger triangle has the same angle θ and sides 24, 70 and 74. Find its $\tan\theta$. What do you notice? Why?

Example 5

Find each side marked x, to the nearest tenth.

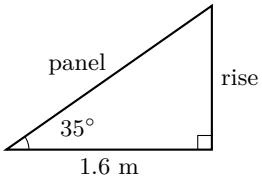






Example 6

A solar panel on a farm near Taber is tilted at 35° to the ground. Seen from the side, the panel covers a horizontal distance of 1.6 m.



(a) How much higher is the top edge of the panel than the bottom edge (the rise)?

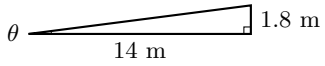
(b) How long is the panel (the sloped side)?

(c) Check (b) with the Pythagorean theorem.

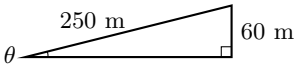
4. Finding Angles, Elevation and Depression

Example 7

(a) A boat launch drops 1.8 m over a horizontal distance of 14 m. Find the angle the launch makes with the horizontal.

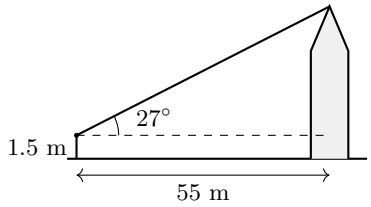


(b) A straight path up the side of a coulee is 250 m long and climbs 60 m. Find the angle between the path and the horizontal, and the other acute angle of the triangle.

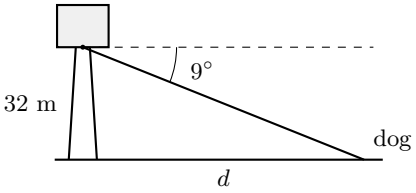


Example 8

(a) Nolan stands 55 m from the base of a church steeple. With a clinometer held at eye level, 1.5 m above the ground, the angle of elevation to the tip of the steeple is 27°. How tall is the steeple?



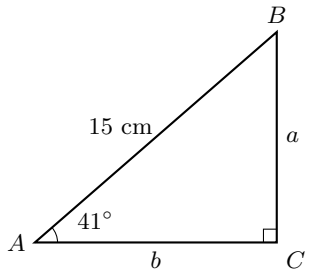
(b) A worker on the walkway of a water tower, 32 m above the ground, sees a lost dog at an angle of depression of 9°. How far is the dog from the base of the tower? (Flat ground; the sketch is not to scale.)



5. Solving Right Triangles

Example 9

In $\triangle ABC$, $\angle C = 90°$, $\angle A = 41°$ and the hypotenuse $c = 15$ cm. Solve the triangle (find $\angle B$, a and b).



Example 10

A cell tower is 52 m tall. Two guy wires run from the top of the tower to anchors on opposite sides of the tower, on flat ground. One wire makes an angle of 61° with the ground and the other makes 48°. How far apart are the two anchors?

